Review notes · for you

Build a circuit: unit check

IoT · Level 2 · Unit check

Materials
Paper and a pen. No device needed.

Learning goals

- Make it light up
- Switches that decide
- Talk in binary
- Meet the microcontroller

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Quiz sheet

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Name _____ Class _____ Date _____

1 **Grace's LED stays dark. The battery is new, the resistor is 100 Ω and the loop is closed. What should she check first?**

◯ That the LED's long leg goes to + ◯ Whether a bigger battery would help ◯ Whether to take out the resistor

2 **Match each problem to its fix.**

An LED burnt out in a flash	• • • •	Change OR to AND
The fan runs in empty rooms	• • • •	Read it with more bits
The light sensor misses small changes	• • • •	Strip the wire end
The night light never comes on at night	• • • •	Add a resistor
The bulb stays dark: a wire end is still in plastic	• • • •	Put the program in a forever loop

3 **A red LED on a 4.5 V battery. Work out the resistor.**

4.5 V minus 2 V for the LED leaves _____ V. For 10 mA, the resistor should be about _____ Ω.

Word bank: 450 · 4.5 · 2.5 · 250

4 **A rule with three inputs has 8 rows in its truth table.**

◯ True ◯ False

5 **The school bell should ring at 10:30, but only on school days. Which rule?**

◯ 10:30 OR school day ◯ 10:30 AND NOT school day ◯ 10:30 AND school day

6 **Which of these fit in one byte?**

☐ The number 300 ☐ One Arabic letter ☐ The letter A ☐ The number 200 ☐ A 10-bit sensor reading

7 **Put these in order of size, smallest first.**

☐ One 10-bit sensor reading

☐ A full Arabic SMS

☐ One bit

☐ A phone photo

☐ A page of a book

8 **Which binary number is 20?**

◯ 00101 ◯ 10010 ◯ 10100

9 **A torch bulb lights if one wire runs from the battery to the bulb's tip, and nothing else touches the bulb.**

◯ True ◯ False

10 **Which are true for a microcontroller night light?**

☐ It sets up once, then loops forever ☐ It works without any program ☐ It reads the light as a number

☐ It needs the internet to decide ☐ You can change its threshold to suit the room

Check your answers

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1. **That the LED's long leg goes to +**

   An LED works only one way round, and backwards is the most common fault.

2. **An LED burnt out in a flash — Add a resistor**
   The fan runs in empty rooms — Change OR to AND
   The light sensor misses small changes — Read it with more bits
   The night light never comes on at night — Put the program in a forever loop
   The bulb stays dark: a wire end is still in plastic — Strip the wire end

   Each fix comes from one lesson: a resistor limits the current, AND needs both inputs, more bits make finer steps, the loop keeps checking, and bare copper closes the loop.

3. **4.5 V minus 2 V for the LED leaves 2.5 V. For 10 mA, the resistor should be about 250 Ω.**

   4.5 − 2 = 2.5 V for the resistor, and 2.5 ÷ 0.01 = 250 Ω.

4. **True**

   Each input can be 0 or 1, so each new input doubles the rows: 2, 4, 8.

5. **10:30 AND school day**

   Both must be true, so it's AND. With OR, it would ring every day at 10:30, and all day long on school days; with AND NOT, it would ring only on days off.

6. **The letter A; The number 200**

   A byte holds 0 to 255, so 200 fits and 300 doesn't.

7. **Top to bottom: 2, 3, 1, 5, 4**

   1 bit; 10 bits (2 bytes); up to 140 bytes; about 2,000 bytes; about 3,000,000 bytes.

8. **10100**

   20 = 16 + 4, so the 16 and 4 places are on: 10100.

9. **False**

   Current needs a closed loop through both of the bulb's contacts, the tip and the metal side, and back to the battery.

10. **It sets up once, then loops forever; It reads the light as a number; You can change its threshold to suit the room**

    Setup, a loop, a reading turned into a number and a threshold you choose: that's every night light.